

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First Semester of B.Pharm (OLD) Examination
University Theory Examination December 2015
MA131 Basic Mathematics

Date: 26/12/2015 Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80
Saturday

Instructions:

- (1) Figures to right indicate marks.
- (2) Section wise separate answer book to be used.
- (3) Draw neat sketches wherever necessary.
- (4) Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Attempt any TWO of the following.

(a) Answer the following questions.

[10]

- 1) Consider the Universal set $S = \{1, 2, \dots, 9\}$ and the sets
 $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 6, 7\}$, $C = \{5, 6, 7, 8, 9\}$,
 $D = \{1, 3, 5, 7, 9\}$, $E = \{2, 4, 6, 8\}$, $F = \{1, 5, 9\}$
Find (i) $A \cup B$ (ii) $B \cap D$ (iii) $A \cap (B \cup E)$ (iv) $A \cap B'$ (v) $D \cap F$
- 2) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$,
define $n(A)$ = number of elements in set A , then verify following.
(i) $n(A') = n(U) - n(A)$
(ii) $n(A \cap B') = n(A) - n(A \cap B)$
(iii) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
(iv) $n(A \cup B)' = n(A' \cap B')$

(b) Answer the following questions.

[10]

- 1) Let $A = \{1, 2\}$, $B = \{a, b, c\}$ and $C = \{c, d\}$,
Determine (i) $(A \times B) \cap (A \times C)$ (ii) $A \times (B \cap C)$
- 2) Let $A = \{1, 2, 3, 4\}$. Determine whether or not each relation below is a function
from A into A
(i) $f = \{(2, 3), (1, 4), (2, 1), (3, 2), (4, 4)\}$
(ii) $g = \{(3, 1), (4, 2), (1, 1)\}$
(iii) $h = \{(2, 1), (3, 4), (1, 4), (2, 1), (4, 4)\}$

(c) Answer the following questions.

[10]

- 1) Let the function $f : R \rightarrow R$ be defined as follows

$$f(x) = \begin{cases} 3x - 1 & \text{if } x > 3 \\ x^2 - 2 & \text{if } -2 \leq x \leq 3 \\ 2x + 3 & \text{if } x < -2 \end{cases}$$

Find (i) $f(2)$ (ii) $f(4)$ (iii) $f(-1)$ (iv) $f(-3)$

2) Sketch the graph of each function:

$$(i) f(x) = \frac{1}{2}x - 1 \quad (ii) g(x) = x^3 - 3x + 2 \quad (iii) h(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1/x & \text{if } x \neq 0 \end{cases}$$

Q.2. Attempt any TWO of the following.

(a) Evaluate the limits (i) $\lim_{x \rightarrow -1} \frac{\sqrt{x^2 + 8} - 3}{x + 1}$ (ii) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^4 - 16}$ [10]

(b) Using limit definition of derivative of function, differentiate functions (i) $f(x) = \frac{x}{x-1}$ (ii) $f(x) = \sqrt{x}, x > 0$ [10]

(c) Answer the following questions [10]

1) If $y = x^2 - \sin x$, find $\frac{dy}{dx}$

3) If $y = 5x + \cos x$ find $\frac{dy}{dx}$

2) Find $\left. \frac{dy}{dx} \right|_{x=\sqrt{3}}$ if $y = 1 - \frac{1}{x}$

4) Find $\left. \frac{ds}{dt} \right|_{t=-1}$ if $s = 1 - 3t^2$

SECTION II

Q.3. Attempt any TWO of the following.

(a) Answer the following questions [10]

1) suppose that f is integrable and that $\int_0^3 f(z) dz = 3$ and $\int_0^4 f(z) dz = 7$ Find

$$(i) \int_3^4 f(z) dz \quad (ii) \int_4^3 f(t) dt$$

2) Evaluate $\int_0^2 (3x^2 + x - 5) dx$

(b) Find the inverse of a matrix D where $D = \begin{bmatrix} 1 & 1 & 6 \\ 2 & 1 & 3 \\ 4 & 2 & 2 \end{bmatrix}$ [10]

Also verify by Matrix multiplication $DD^{-1} = I$

(c) If $A = \begin{bmatrix} 2 & 9 & 1 \\ 6 & 0 & 3 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ [10]
Then verify $|AB| = |A| \times |B|$

Q.4. Attempt any TWO of the following.

(a) Consider the matrices [10]

$$A = \begin{bmatrix} 0 & 5 & 2 \\ 5 & 6 & 4 \\ 2 & 4 & -7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 3 & 4 \\ -3 & 1 & -7 \\ -4 & 7 & 2 \end{bmatrix}$$

Find

- 1) Matrix product BA
- 2) Verify $(A + 2B)' = A' + 2B'$

(b) Answer the following questions [10]

- 1) Data on the percentage of paracetamol in batches of Sterwin 500 mg tablets are shown below (Table 1). The tablets were assayed by both the BP1993 UV method and by an alternative NIR reflectance spectrophotometric method.

Table 1: Paracetamol content (%) in batches of tablets

Batch	1	2	3	4	5	6	7	8	9	10
UV	84.06	84.1	85.13	84.96	84.96	84.87	84.75	83.93	83.92	84.39

Compute Arithmetic Mean and Standard deviation percentage of paracetamol respectively.

- 2) Solve using Cramer's Rule $3x + 5y = 7, 2x - 6y = 4$

(c) Answer the following questions [10]

- 1) Pharmaceutical company supplies batches of tablets to a customer. The QC laboratories of both companies perform extensive analysis on samples from each batch. For many months the results from the two laboratories had been unsatisfactory in their lack of agreement. Table 2 shows results for one parameter, the one-hour dissolution rate.

Table 2: One-hour dissolution rate

30.7	28.9	41.2	37.7	29.1	35.0	48.1	34.4	35.3	31.7
38.2	30.0	31.4	41.2	35.6	37.0	31.5	41.8	33.5	26.5
42.1	31.6	39.7	49.3	30.6	45.8	34.0	27.5	40.9	44.7
34.4	48.0	37.1	30.9	34.2	41.2	34.9	35.8	30.7	26.8
37.1	29.1	27.5	35.6	39.5	40.9	31.6	31.8	26.7	38.2

Table 3: Class Limits

Lower Limit	26.0	30.0	34.0	38.0	42.0	46.0
Upper Limit	29.9	33.9	37.9	41.9	45.9	49.9

Construct the frequency distribution based on the Table 2. Use class limits in Table 3

2) Solve the following simultaneous equations using Cramer's rule.

$$x + y + z = 4$$

$$2x - 3y + 4z = 33$$

$$3x - 2y - 2z = 2$$